

## LIST OF EXPERIMENTS

1. To determine the melting point of a substance using a melting point apparatus.
2. To determine the boiling point of a liquid using a boiling point apparatus.
3. To prepare a standard solution of a given solute and determine its concentration using titration.
4. To perform a simple distillation of a mixture of two miscible liquids and determine their relative amounts.
5. To perform a fractional distillation of a mixture of two miscible liquids and determine their relative amounts.
6. To determine the solubility of a solute in a solvent at different temperatures and plot a solubility curve.
7. To prepare a colloidal solution of a given solute and study its properties.
8. To study the effect of concentration on the rate of reaction between sodium thiosulphate and hydrochloric acid.
9. To study the effect of temperature on the rate of reaction between potassium permanganate and oxalic acid.
10. To study the effect of catalyst on the rate of reaction between hydrogen peroxide and potassium iodide.
11. To determine the empirical formula of a compound by the method of combustion.
12. To determine the molecular weight of a volatile liquid by Victor Meyer's method.
13. To determine the surface tension of a liquid by the capillary rise method.
14. To determine the refractive index of a glass prism using a spectrometer.
15. To determine the focal length of a convex lens using a lens formula.

These experiments are essential for gaining practical knowledge in the field of chemistry. They help in understanding the concepts of chemistry in a better way and provide hands-on experience in performing various laboratory techniques. It is important to perform these experiments under the guidance of a trained professional in a well-equipped laboratory to ensure safety and accuracy of the results.

**Note: A minimum of ten experiments from the following should be performed.**

1. Acid-Base Titrations: This experiment involves the determination of the concentration of an acid or base solution using a titration technique.
2. Solubility Experiments: Solubility experiments are used to determine the solubility of a substance in a particular solvent at a given temperature.
3. Spectrophotometry: This experiment uses spectrophotometers to measure the absorbance of light by a sample to determine the concentration of a solution.

4. Chromatography: Chromatography is a technique used to separate the components of a mixture based on their different physical and chemical properties.
5. Electrochemistry: This experiment involves the study of the relationship between electricity and chemical reactions.
6. Calorimetry: Calorimetry experiments are used to determine the heat of a reaction or the heat capacity of a substance.
7. Reaction Kinetics: This experiment involves the study of the rate of a chemical reaction and the factors that affect it.
8. Gas Laws: Gas laws experiments are used to study the behavior of gases under different conditions of temperature, pressure, and volume.
9. Acid-Base Properties: This experiment involves the determination of the acid-base properties of different substances.
10. Redox Reactions: Redox reactions involve the transfer of electrons between reactants and are studied using various techniques such as titrations and electrochemistry.
11. Polymerization: Polymerization experiments involve the synthesis of polymers from monomers, and the characterization of the resulting polymers.
12. Thermodynamics: Thermodynamics experiments involve the study of energy and its transformations in chemical reactions.

Performing these experiments will help you gain a deeper understanding of fundamental concepts in chemistry and develop key laboratory skills necessary for a career in the field.

#### **(A) Hardware based experiments**

- Hardware experiments involve the use of physical components to test and verify the functionality of a system or device.
- These experiments can range from simple circuit testing to complex system integration and testing.
- Hardware experiments are often used in the development of new products or to troubleshoot existing systems.
- One common type of hardware experiment is circuit testing, which involves the use of a circuit board to test the functionality of individual components and their interactions with each other.
- Another type of hardware experiment is system integration testing, which involves the use of multiple hardware components to test the functionality of a larger system.
- Hardware experiments can also be used to test the durability and reliability of a system, such as through stress testing or environmental testing.

- It is important to carefully plan and execute hardware experiments, as they can be expensive and time-consuming.
- Proper safety precautions should always be taken when conducting hardware experiments, such as using protective equipment and following established protocols.

## Verification of Kirchhoff's laws

Kirchhoff's laws are fundamental laws in the study of electrical circuits that help in determining the values of voltage, current, and resistance in a circuit. There are two laws, Kirchhoff's Current Law (KCL) and Kirchhoff's Voltage Law (KVL), which are used to analyze complex circuits. Verification of these laws is crucial to ensure accurate analysis of circuits. Here are the ways in which Kirchhoff's laws can be verified:

1. Verification of Kirchhoff's Current Law (KCL)
  - KCL states that the algebraic sum of currents flowing into a node is equal to the algebraic sum of currents flowing out of that node.
  - To verify KCL, select a node in the circuit and label the currents flowing into the node as positive and the currents flowing out of the node as negative.
  - Sum up all the currents at that node and check if the sum is equal to zero. If the sum is zero, KCL is verified.
2. Verification of Kirchhoff's Voltage Law (KVL)
  - KVL states that the algebraic sum of voltages around any closed loop in a circuit is equal to zero.
  - To verify KVL, select a closed loop in the circuit and assign polarities to the voltages on the loop.
  - Starting from any point on the loop, traverse the loop in a clockwise or anticlockwise direction and add up all the voltages. The sum of voltages should be equal to zero if KVL is verified.
3. Verification of Kirchhoff's laws using circuit simulators
  - Circuit simulators like SPICE or LTSpice can be used to verify Kirchhoff's laws.
  - Build the circuit in the simulator and run a simulation to obtain the values of voltage and current in the circuit.
  - Verify KCL and KVL by applying the laws to the simulated values and checking if they are satisfied.

In conclusion, Kirchhoff's laws are fundamental laws in the study of electrical circuits, and verifying them is crucial to ensure accurate analysis of circuits. The methods mentioned above can be used to verify KCL and KVL in a circuit.

## **Measurement of Power and Power Factor in a Single Phase AC Series Inductive Circuit and Study Improvement of Power Factor Using Capacitor**

In the study of electrical circuits, the measurement of power and power factor is crucial. In a single phase AC circuit, the power factor is defined as the ratio of the real power, which is the power that is actually used in the circuit, to the apparent power, which is the product of the voltage and current. In this topic, we will learn about the measurement of power and power factor in a single phase AC series inductive circuit, and how the power factor can be improved using a capacitor.

### **Measurement of Power**

The power in a single phase AC circuit can be measured using a wattmeter. The wattmeter consists of two coils, one of which is connected in series with the load, and the other is connected in parallel with the load. The voltage coil is connected in parallel with the load, while the current coil is connected in series with the load. The wattmeter measures the product of the voltage and current, which gives the power consumed by the load.

### **Measurement of Power Factor**

The power factor of a single phase AC circuit can be measured using a power factor meter. The power factor meter consists of a fixed coil and a moving coil. The fixed coil is connected in series with the load, while the moving coil is connected across the load. The power factor meter measures the angle between the voltage and current, which gives the power factor of the circuit.

### **Single Phase AC Series Inductive Circuit**

In a single phase AC series inductive circuit, the power factor is usually low due to the presence of inductance. Inductance causes the current to lag behind the voltage, which results in a low power factor. The power factor can be improved by adding a capacitor in parallel with the load. The capacitor supplies reactive power to the circuit, which helps to reduce the lagging current and improve the power factor.

### **Improvement of Power Factor Using Capacitor**

The power factor of a single phase AC series inductive circuit can be improved by adding a capacitor in parallel with the load. The value of the capacitor should be chosen so that it supplies the reactive power required to compensate for the lagging current. The reactive power can be calculated using the formula:

$$Q = V^2 / (L)$$

where  $Q$  is the reactive power,  $V$  is the voltage,  $\omega$  is the angular frequency of the circuit, and  $L$  is the inductance.

By adding a capacitor of the appropriate value in parallel with the load, the power factor can be improved, which results in a more efficient use of the electrical energy. The improvement in power factor can be measured using a power factor meter, which will show a higher power factor after the capacitor is added.

To summarize, the measurement of power and power factor in a single phase AC series inductive circuit is important for the efficient use of electrical energy. The power factor can be improved by adding a capacitor in parallel with the load, which supplies reactive power to the circuit and reduces the lagging current.

### **Study of phenomenon of resonance in RLC series circuit and obtain resonant frequency.**

In this topic, we will be studying the phenomenon of resonance in an RLC series circuit and how to obtain the resonant frequency. The RLC series circuit consists of a resistor ( $R$ ), an inductor ( $L$ ), and a capacitor ( $C$ ) connected in series.

The following points will help you understand the phenomenon of resonance in an RLC series circuit:

- Resonance is a phenomenon that occurs when an alternating current (AC) is passed through an RLC series circuit and the circuit reaches a state where the voltage across the capacitor and the current flowing through the circuit are in phase.
- At resonance, the impedance of the circuit is minimum and the current flowing through the circuit is maximum.
- The resonant frequency of an RLC series circuit can be calculated using the formula:

$$f = 1 / (2 \sqrt{LC})$$

where  $f$  is the resonant frequency in Hertz (Hz),  $L$  is the inductance in Henrys (H), and  $C$  is the capacitance in Farads (F).

- The resonant frequency of an RLC series circuit can also be obtained experimentally by varying the frequency of the AC source and measuring the current flowing through the circuit using an ammeter.
- The resonant frequency can be identified by the frequency at which the current flowing through the circuit is maximum.
- The quality factor ( $Q$ ) of an RLC series circuit is a measure of the sharpness of resonance and is defined as the ratio of the energy stored in the circuit to the energy dissipated in the circuit.

$$Q = 1 / R \sqrt{L / C}$$

where  $R$  is the resistance in Ohms ( $\Omega$ ),  $L$  is the inductance in Henrys (H), and  $C$  is the capacitance in Farads (F).

- The quality factor of an RLC series circuit can be measured experimentally by measuring the bandwidth of the circuit at half-power points.

In conclusion, the study of the phenomenon of resonance in an RLC series circuit and obtaining the resonant frequency is important in understanding the behavior of electrical circuits. The resonant frequency can be calculated using a formula or obtained experimentally, and the quality factor of the circuit can be measured to determine the sharpness of resonance.

### **Connection and Measurement of Power Consumption of a Fluorescent Lamp (Tube Light)**

Fluorescent lamps, also known as tube lights, are commonly used as a source of illumination in homes, offices, and other commercial buildings. These lamps are known for their energy efficiency and long lifespan, making them an ideal choice for lighting solutions.

Here are some important points to consider when connecting and measuring the power consumption of a fluorescent lamp:

1. Connection of Fluorescent Lamp:
  - Before connecting the fluorescent lamp, ensure that the power supply is turned off.
  - Check the voltage and current rating of the fluorescent lamp and ensure that they match the requirements of the power supply.
  - Connect the lamp to the power supply using the appropriate wiring and connectors.
  - Turn on the power supply and check if the lamp is functioning properly.
2. Measurement of Power Consumption:
  - To measure the power consumption of a fluorescent lamp, use a wattmeter or a power analyzer.
  - Connect the wattmeter or power analyzer to the power supply and the fluorescent lamp.
  - Turn on the power supply and monitor the power consumption displayed on the wattmeter or power analyzer.
  - Calculate the power consumption of the fluorescent lamp by multiplying the voltage and current readings obtained from the wattmeter or power analyzer.
3. Factors Affecting Power Consumption:
  - The power consumption of a fluorescent lamp is affected by various factors such as the voltage and current rating, the length of the lamp, and the type of ballast used.

- Using a ballast that is not compatible with the fluorescent lamp can result in a higher power consumption and reduced lifespan of the lamp.
- It is important to ensure that the ballast is correctly matched with the fluorescent lamp to optimize its performance and reduce power consumption.

In conclusion, connecting and measuring the power consumption of a fluorescent lamp requires careful attention to detail and adherence to safety protocols. By following the above points, you can ensure that the fluorescent lamp operates efficiently and with minimal power consumption.

### **Introduction**

- The measurement of power in a three-phase circuit is essential for the proper functioning of various electrical systems.
- The two-wattmeter method is commonly used to measure power in three-phase circuits.
- This method involves using two wattmeters to measure the power in each of the three phases and then calculating the total power.

### **Two-wattmeter method**

- The two-wattmeter method involves connecting two wattmeters in the circuit.
- One wattmeter is connected between the first phase and the second phase, while the other wattmeter is connected between the second phase and the third phase.
- The voltage across each phase and the current through each phase are measured using voltmeters and ammeters, respectively.
- The wattmeters measure the power in each phase, and the total power is calculated by adding the two readings.

### **Power factor for star-connected load**

- The power factor for a star-connected load can be determined using the two-wattmeter method.
- The power factor is given by the formula  $\cos(\Phi) = (P_1 - P_2) / 3V_L I \cos(\Phi)$ , where  $P_1$  and  $P_2$  are the readings of the two wattmeters,  $V$  is the line voltage,  $I$  is the line current, and  $\Phi$  is the phase angle between voltage and current.

### **Power factor for delta-connected load**

- The power factor for a delta-connected load can also be determined using the two-wattmeter method.
- The power factor is given by the formula  $\cos(\Phi) = (P_1 - P_2) / 3V_L I \cos(\Phi)$ , where  $P_1$  and  $P_2$  are the readings of the two wattmeters,  $V$  is the phase

voltage,  $V_L$  is the line voltage,  $I_L$  is the line current, and  $\phi$  is the phase angle between voltage and current.

## Conclusion

- The two-wattmeter method is an effective way to measure power in three-phase circuits.
- By using this method, the power factor for both star and delta-connected loads can also be determined.
- Proper measurement of power and power factor is essential for efficient operation of electrical systems.

## 6. Determination of parameters of ac single phase series RLC circuit.

In order to determine the parameters of an AC single phase series RLC circuit, the following steps need to be followed:

1. Measure the resistance ( $R$ ) of the circuit using a multimeter. This can be done by connecting the multimeter in series with the circuit and taking the reading.
2. Measure the inductance ( $L$ ) of the circuit using an inductance meter. This can be done by connecting the inductance meter in series with the circuit and taking the reading.
3. Measure the capacitance ( $C$ ) of the circuit using a capacitance meter. This can be done by connecting the capacitance meter in parallel with the circuit and taking the reading.

4. Calculate the resonant frequency ( $f_r$ ) of the circuit using the formula:

$$f_r = 1 / (2 * \pi * \sqrt{L * C})$$

where  $\pi$  is the mathematical constant and  $\sqrt{\phantom{x}}$  is the square root function.

5. Apply a sinusoidal voltage to the circuit and measure the current ( $I$ ) and the voltage ( $V$ ) across the circuit using an oscilloscope.
6. Calculate the impedance ( $Z$ ) of the circuit using the formula:

$$Z = V / I$$

7. Calculate the phase angle ( $\phi$ ) between the voltage and current using the formula:

$$\phi = \arctan((\omega L - 1/\omega C) / R)$$

where  $\arctan$  is the inverse tangent function and  $\omega$  is the angular frequency.

8. Calculate the reactance ( $X$ ) of the circuit using the formula:

$$X = Z * \sin(\phi)$$



9. Calculate the inductive reactance ( $X_L$ ) using the formula:

$$X_L = \omega L$$

10. Calculate the capacitive reactance ( $X_C$ ) using the formula:

$$X_C = 1 / (\omega C)$$

11. Calculate the quality factor ( $Q$ ) of the circuit using the formula:

$$Q = X_L / R$$

12. Calculate the bandwidth ( $BW$ ) of the circuit using the formula:

$$BW = f_r / Q$$

where  $f_r$  is the resonant frequency calculated in step 4.

By following these steps, the parameters of an AC single phase series RLC circuit can be determined. These parameters are important for designing and analyzing circuits in various applications.

#### **7. Determination of (i) Voltage ratio (ii) polarity and (iii) efficiency by load test of a single phase Transformer.**

1. Load test is an important test conducted on a single phase transformer to determine various parameters such as voltage ratio, polarity, and efficiency.
2. Voltage ratio is the ratio of the output voltage to the input voltage of the transformer.
3. The load test involves connecting the primary winding of the transformer to a source of alternating current and the secondary winding is connected to a load.
4. The load is gradually increased, and the corresponding values of input current, output current, and voltage are measured.
5. By measuring the input and output voltage and current, the voltage ratio of the transformer can be calculated.
6. Polarity of a transformer refers to the direction of the induced voltage in the secondary winding with respect to the primary winding.
7. To determine the polarity of the transformer, a low voltage battery is connected to the primary winding, and the direction of the induced voltage in the secondary winding is observed.
8. If the induced voltage is in the same direction as the battery voltage, then the polarity is said to be additive. If the induced voltage is in the opposite direction, then the polarity is said to be subtractive.
9. Efficiency of a transformer is defined as the ratio of output power to input power.

10. The efficiency of the transformer can be calculated by measuring the input and output power and then taking the ratio of the two.
11. The load test is an essential test for determining the various parameters of a single phase transformer, and it is essential to conduct this test to ensure the proper functioning of the transformer.

### **8. Determination of efficiency of a DC shunt motor by load test**

In order to determine the efficiency of a DC shunt motor, a load test is performed. The load test is carried out by applying a mechanical load to the motor and measuring the corresponding electrical parameters.

The following points explain the procedure for determining the efficiency of a DC shunt motor by load test:

1. Firstly, the motor is connected to a DC power supply and its rated voltage is applied.
2. The speed of the motor is measured using a tachometer.
3. A suitable load is applied to the motor and the current drawn by the motor is measured using an ammeter.
4. The voltage across the motor terminals is measured using a voltmeter.
5. The power input to the motor is calculated using the formula  $P = VI$ , where  $P$  is the power input,  $V$  is the voltage across the motor terminals, and  $I$  is the current drawn by the motor.
6. The output power of the motor is calculated using the formula  $P_{out} = T \omega$ , where  $P_{out}$  is the output power,  $T$  is the torque developed by the motor, and  $\omega$  is the angular speed of the motor.
7. The efficiency of the motor is calculated using the formula  $\eta = P_{out}/P_{in} \times 100\%$ , where  $\eta$  is the efficiency of the motor,  $P_{out}$  is the output power of the motor, and  $P_{in}$  is the power input to the motor.
8. The load is varied and the above steps are repeated to obtain efficiency values at different loads.
9. A graph is plotted with load on the x-axis and efficiency on the y-axis.
10. The maximum efficiency point is determined from the graph.
11. The efficiency of the motor at the maximum efficiency point is noted as the overall efficiency of the motor.

By performing a load test on a DC shunt motor, its efficiency can be determined accurately. This information can be used for selecting the appropriate motor for a specific application or for optimizing the performance of an existing motor.

### **9. To study running and speed reversal of a three phase induction motor and record speed in both directions.**

In order to understand the running and speed reversal of a three phase induction motor, it is important to follow the guidelines mentioned below:

- Firstly, connect a three phase induction motor to a three phase AC supply. Ensure that the motor is properly grounded.
- Start the motor and measure the initial speed using a tachometer. Record this value.
- Now, reverse the direction of the motor by interchanging any two of the three supply phase wires. The motor should now run in the opposite direction.
- Measure the speed of the motor again using the tachometer. Record this value as well.
- Repeat the above steps multiple times to get an average value for both the forward and reverse directions.
- Analyze the data collected to understand the running characteristics and speed reversal of the three phase induction motor.
- Make a note of any differences observed in the motor's performance while running in the forward and reverse directions.
- Also, study the effects of changing the supply frequency and voltage on the motor's performance.
- Finally, draw conclusions based on the data collected and observations made during the experiment.

By following the above guidelines and thoroughly analyzing the data collected, one can gain a clear understanding of the running and speed reversal of a three phase induction motor.

### **10. Demonstration of cut-out sections of machines: dc machine, three phase induction machine, single-phase induction machine and synchronous machine.**

Here are some key points to keep in mind while demonstrating cut-out sections of machines:

- Cut-out sections of machines are used to study the internal components and working of machines.
- DC machines consist of a stator and rotor. The stator houses the field winding while the rotor houses the armature winding. The cut-out section of a DC machine enables the study of the commutator, brushes, armature winding, and field winding.
- Three-phase induction machines consist of a stator and rotor. The stator houses the three-phase windings while the rotor consists of conductors. The cut-out section of a three-phase induction machine enables the study of the stator, rotor, and windings.

- Single-phase induction machines consist of a stator and rotor. The stator houses the main and auxiliary windings while the rotor consists of a squirrel-cage. The cut-out section of a single-phase induction machine enables the study of the stator, rotor, and windings.
- Synchronous machines consist of a stator and rotor. The stator houses the three-phase windings while the rotor consists of a field winding. The cut-out section of a synchronous machine enables the study of the rotor, stator, and field winding.

It is important to note that while demonstrating cut-out sections of machines, safety precautions should be taken to avoid any accidents or injuries. It is also recommended to have an expert present during the demonstration to guide and supervise the process.

### **(B) Experiments available on virtual lab**

Virtual labs offer a wide range of experiments that can be conducted online. These experiments are designed to simulate real-life experiments, allowing students to gain practical knowledge and experience without the need for a physical laboratory. Some of the experiments available on virtual labs include:

- Biology experiments: Virtual labs offer a range of biology experiments, including the study of cells, tissues, and organs. These experiments allow students to understand the principles of cell biology, genetics, and physiology.
- Chemistry experiments: Virtual labs offer a range of chemistry experiments, including the study of chemical reactions, the properties of elements, and the behavior of gases. These experiments allow students to understand the principles of chemistry and apply them to real-world situations.
- Physics experiments: Virtual labs offer a range of physics experiments, including the study of mechanics, optics, and electromagnetism. These experiments allow students to understand the principles of physics and apply them to real-world situations.
- Engineering experiments: Virtual labs offer a range of engineering experiments, including the study of circuits, robotics, and materials science. These experiments allow students to understand the principles of engineering and apply them to real-world situations.
- Environmental science experiments: Virtual labs offer a range of environmental science experiments, including the study of ecosystems, climate change, and pollution. These experiments allow students to understand the principles of environmental science and apply them to real-world situations.

In conclusion, virtual labs offer a wide range of experiments that can be con-

ducted online without the need for a physical laboratory. These experiments provide students with practical knowledge and experience, allowing them to understand the principles of science and engineering and apply them to real-world situations.

## Kirchhoff's Laws

Kirchhoff's laws are fundamental laws in electrical engineering that are used to analyze and solve electrical circuits. They are named after Gustav Kirchhoff, a German physicist who first introduced them in 1845.

### Kirchhoff's First Law

Also known as the law of conservation of charge, Kirchhoff's first law states that the total current flowing into a node or junction in an electrical circuit must be equal to the total current flowing out of that node or junction. In other words, the sum of the currents entering a node must be equal to the sum of the currents leaving that node.

Mathematically, Kirchhoff's first law can be expressed as:

$$\Sigma I(\text{in}) = \Sigma I(\text{out})$$

where  $\Sigma I(\text{in})$  is the sum of the currents entering the node and  $\Sigma I(\text{out})$  is the sum of the currents leaving the node.

### Kirchhoff's Second Law

Also known as the law of conservation of energy, Kirchhoff's second law states that the sum of the voltage drops around any closed loop in an electrical circuit must be equal to the sum of the voltage sources in that loop. In other words, the total energy supplied by the voltage sources in a circuit must be equal to the total energy consumed by the circuit's components.

Mathematically, Kirchhoff's second law can be expressed as:

$$\Sigma V = \Sigma IR$$

where  $\Sigma V$  is the sum of the voltage sources in the loop,  $\Sigma IR$  is the sum of the voltage drops across the resistors in the loop, and  $I$  is the current flowing through the loop.

### How to Apply Kirchhoff's Laws

To apply Kirchhoff's laws to an electrical circuit, you must first identify the nodes and loops in the circuit. A node is any point in the circuit where two or more components are connected, while a loop is any closed path in the circuit that does not pass through any node more than once.

Once you have identified the nodes and loops in the circuit, you can use Kirchhoff's first law to write equations for the currents flowing into and out of each node, and Kirchhoff's second law to write equations for the voltage drops around each loop. You can then solve these equations simultaneously to determine the values of the currents and voltages in the circuit.

## Conclusion

In summary, Kirchhoff's laws are fundamental laws in electrical engineering that are used to analyze and solve electrical circuits. Kirchhoff's first law states that the total current flowing into a node must be equal to the total current flowing out of that node, while Kirchhoff's second law states that the sum of the voltage drops around any closed loop in a circuit must be equal to the sum of the voltage sources in that loop. By applying these laws, you can analyze and solve complex electrical circuits.

## Virtual Lab: A Comprehensive Guide

Virtual Lab is an online platform that provides students with a realistic laboratory experience. Here are some key features of the Virtual Lab:

- **Accessible:** The Virtual Lab is available online and can be accessed from anywhere at any time. All you need is an internet connection.
- **User-friendly:** The platform is easy to navigate and use. You can choose from a wide range of experiments and simulations.
- **Realistic simulations:** The Virtual Lab provides realistic simulations that mimic real-world laboratory experiments. You can perform experiments and analyze data just like you would in a real laboratory.
- **Wide range of experiments:** There are a wide range of experiments available on the Virtual Lab, covering topics such as physics, chemistry, biology, and engineering.
- **Customizable:** The Virtual Lab allows you to customize experiments to suit your needs. You can change variables and parameters to see how they affect the outcome of the experiment.
- **Interactive:** The Virtual Lab is interactive, allowing you to manipulate objects and observe how they behave. You can also interact with other students and instructors in real-time.
- **Cost-effective:** The Virtual Lab is a cost-effective alternative to traditional laboratory experiments. You don't need to purchase expensive equipment or materials, and there are no safety hazards to worry about.
- **Learning resources:** The Virtual Lab provides a range of learning resources, including videos, tutorials, and quizzes. These resources can help you understand the underlying concepts behind the experiments.

In conclusion, the Virtual Lab is a valuable tool for students who want to gain practical laboratory experience from the comfort of their own home. With its wide range of experiments, realistic simulations, and interactive features, the Virtual Lab is an excellent resource for learning and development.

## **Thevenin Theorem**

The Thevenin Theorem is a fundamental concept in electrical engineering that simplifies complex circuits by reducing them to a single voltage source and a single resistor. Here are some key points to understand about the theorem:

- The theorem states that any linear circuit can be reduced to an equivalent circuit that consists of a single voltage source and a single resistor.
- The equivalent circuit will produce the same current through any load as the original circuit would have produced.
- To determine the Thevenin equivalent circuit, we need to calculate the open-circuit voltage at the output terminals and the equivalent resistance of the circuit as seen from those terminals.
- The open-circuit voltage is the voltage that would be measured at the output terminals if the load resistance was infinite.
- The equivalent resistance is the resistance that would be seen by the load if it was connected to the output terminals.
- The Thevenin equivalent circuit can be used to calculate the power dissipated by the load, the maximum power that can be delivered to the load, and the efficiency of the circuit.
- The theorem is particularly useful in circuit analysis because it allows us to simplify complex circuits and reduce the number of calculations required.
- However, the theorem is only applicable to linear circuits, which means that it cannot be used for circuits that contain non-linear components such as diodes or transistors.

In conclusion, the Thevenin Theorem is a powerful tool for simplifying complex circuits and reducing the number of calculations required. By understanding the key concepts of the theorem, engineers can analyze and design more efficient and effective electrical systems.

## **Introduction to Virtual Labs**

Virtual Labs are online platforms that allow students to perform experiments and simulations in a virtual environment. It is a combination of software and hardware that provides students with a virtual laboratory experience. This technology has revolutionized the way students learn by providing them with an interactive and engaging learning environment. Here are some key points about Virtual Labs:

- Virtual Labs provide students with the opportunity to conduct experiments in a safe and controlled environment. This eliminates the risks associated with traditional laboratory experiments.
- Virtual Labs are accessible to students from anywhere in the world, as long as they have an internet connection. This makes it possible for students to learn and conduct experiments at their own pace and on their own schedule.
- Virtual Labs are cost-effective, as they eliminate the need for physical laboratory equipment and supplies. This makes it possible for educational institutions to offer more courses and experiments without incurring additional costs.
- Virtual Labs are designed to be user-friendly and easy to navigate. This makes it possible for students to focus on the learning experience, rather than the technology itself.
- Virtual Labs provide students with immediate feedback and results. This allows students to learn from their mistakes and make adjustments in real-time.
- Virtual Labs are an excellent tool for educators to enhance the learning experience of their students. They can be used as a supplement to traditional classroom instruction or as a standalone learning tool.

## Conclusion

Virtual Labs provide students with a safe, cost-effective, and engaging way to learn and conduct experiments. This technology has the potential to revolutionize the way we teach and learn by providing students with an interactive and immersive learning experience. As Virtual Labs continue to evolve, we can expect to see even more innovative and exciting applications in the field of education.

## 3. RLC Series Resonance

RLC series resonance is a phenomenon that occurs in electrical circuits consisting of a resistor, an inductor, and a capacitor connected in series. In RLC series resonance, the circuit's impedance is at its minimum, allowing for the maximum flow of current through the circuit.

Here are some key points to keep in mind about RLC series resonance:

- RLC series resonance occurs when the inductive and capacitive reactances cancel each other out, leaving only the resistance of the circuit to impede the flow of current.
- The frequency at which RLC series resonance occurs can be calculated using the formula  $f = 1/(2\sqrt{LC})$ , where  $f$  is the resonant frequency,  $L$  is the inductance of the circuit, and  $C$  is the capacitance of the circuit.
- At the resonant frequency, the voltage across the capacitor and inductor is equal, and the current through the circuit is in phase with the voltage.



- RLC series resonance can be used to create bandpass filters, which allow a specific range of frequencies to pass through the circuit while blocking others.
- In practical applications, RLC series resonance can be used in radio tuning circuits, audio equalizers, and power factor correction circuits.

It is important to have a clear understanding of RLC series resonance and its applications in electrical circuits, as it is a fundamental concept in electrical engineering and can be a useful tool in designing and analyzing circuits.

### **Virtual Lab**

Virtual Lab is an online platform that allows students to conduct experiments in a simulated environment. It provides a safe and cost-effective way for students to learn and practice different experiments without requiring any physical equipment.

Here are some key features of Virtual Lab:

- **Wide range of experiments:** Virtual Lab offers a wide range of experiments from various disciplines such as physics, chemistry, biology, and engineering. Students can choose experiments based on their interests and learning goals.
- **Simulated environment:** The experiments are conducted in a simulated environment that closely resembles the real-world environment. This allows students to practice the experiments in a safe and controlled environment.
- **Interactive interface:** The interface of Virtual Lab is interactive and user-friendly. It allows students to easily navigate through different experiments and perform various actions such as changing parameters and observing results.
- **Real-time feedback:** Virtual Lab provides real-time feedback on the experiments. This allows students to understand the concepts better and make necessary adjustments to their experiments.
- **Cost-effective:** Virtual Lab is a cost-effective way for students to learn and practice experiments. It eliminates the need for expensive equipment and reduces the cost of conducting experiments.

In conclusion, Virtual Lab is a useful platform for students to learn and practice experiments in a safe and cost-effective way. It provides a wide range of experiments, a simulated environment, an interactive interface, real-time feedback, and cost-effectiveness.

### **4. Measurement of power in 3-phase circuit by two wattmeter method and determination of its power factor for star as well as delta connected load.**

In a three-phase circuit, power can be measured by using two wattmeters. This method is called the two wattmeter method. In this method, two wattmeters are connected in the circuit, and the readings of these wattmeters are used to calculate the power.

The power factor of the load can also be determined using the two wattmeter method. The power factor is an important parameter that describes the efficiency of the circuit.

The following are the steps for measuring power in a three-phase circuit using the two wattmeter method:

1. Connect the two wattmeters in the circuit. One wattmeter is connected between the first phase and the second phase, and the other wattmeter is connected between the second phase and the third phase.
2. Note down the readings of the two wattmeters.
3. Calculate the total power by adding the readings of the two wattmeters. The formula for calculating the total power is  $P = W_1 + W_2$ .
4. Calculate the power factor of the circuit using the formula  $PF = P/S$ , where  $P$  is the total power, and  $S$  is the apparent power.
5. The above steps can be repeated for both star and delta connected loads.

In conclusion, the two wattmeter method is an effective way to measure power in a three-phase circuit. It is also useful for determining the power factor of the load.

### **Introduction to Virtual Lab for Measurement**

Virtual labs are online educational tools that allow students to perform experiments and simulations in a virtual environment. The virtual lab link <http://vp-dei.vlabs.ac.in/Dreamweaver/measurement.html> is a platform that provides students with a unique opportunity to learn about measurement techniques and principles in a safe, controlled environment.

Here are some key points to remember when using the virtual lab link for measurement:

- The virtual lab link provides a range of measurement tools, including rulers, protractors, and micrometers, to help students learn about the different measurement techniques used in science and engineering.
- The lab link also provides interactive simulations that allow students to explore the principles of measurement, such as accuracy, precision, and error analysis.
- The virtual lab link is designed to be user-friendly, with easy-to-use interfaces and clear instructions to guide students through each experiment and simulation.

- Students can access the virtual lab link from anywhere with an internet connection, making it a convenient and flexible learning tool.
- The virtual lab link is ideal for students who may not have access to traditional laboratory facilities, or for those who prefer to learn at their own pace and on their own schedule.
- The virtual lab link provides a safe and controlled environment for students to learn about measurement techniques, without the risks and hazards associated with traditional laboratory experiments.

In conclusion, the virtual lab link for measurement is a valuable tool for students who want to learn about measurement techniques and principles in a safe and controlled environment. By using the virtual lab link, students can gain a deeper understanding of the principles of measurement, and develop the skills and knowledge they need to succeed in science and engineering.

### **Determination of Parameters of AC Single Phase Series RLC Circuit:**

In an AC single-phase series circuit, the circuit components include a resistor, an inductor, and a capacitor connected in series. The circuit's parameters, such as impedance, phase angle, and power factor, can be determined using the following methods:

1. Impedance: The impedance of the circuit can be determined by measuring the voltage across the circuit and the current flowing through it using an oscilloscope. The impedance is given by the following formula:

$$Z = V / I$$

2. Phase Angle: The phase angle between the voltage and current can be determined by measuring the time difference between the voltage and current waveforms on the oscilloscope. The phase angle is given by the following formula:

$$\Phi = \cos^{-1} (P.F)$$

3. Power Factor: The power factor of the circuit can be determined by measuring the active power (P) and the apparent power (S) using a wattmeter. The power factor is given by the following formula:

$$P.F = P / S$$

4. Resonant Frequency: The resonant frequency of the circuit can be determined by measuring the frequency at which the impedance is maximum using an LCR meter. The resonant frequency is given by the following formula:

$$f = 1 / (2 \sqrt{LC})$$

5. Quality Factor: The quality factor of the circuit can be determined by measuring the ratio of energy stored in the circuit to the energy dissipated

per cycle. The quality factor is given by the following formula:

$$Q = 1 / R \sqrt{L / C}$$

Overall, the determination of parameters of an AC single-phase series RLC circuit requires the use of various equipment and formulas. Understanding these parameters and their calculations is crucial in the analysis and design of electrical circuits.

### **Introduction to Virtual Lab**

Virtual Lab is a platform that allows students to conduct experiments and simulations in a virtual environment. Here are some key points you should know about Virtual Lab:

- Virtual Lab allows students to experiment with real-world scenarios, without the need for a physical lab or equipment.
- The platform offers a range of experiments and simulations in various subjects, including physics, chemistry, biology, and engineering.
- Virtual Lab provides a safe and controlled environment for students to perform experiments, as they can make mistakes and learn from them without any risk of harm or damage.
- The experiments and simulations are interactive and engaging, with visual aids and step-by-step instructions to guide students through the process.
- Virtual Lab also offers quizzes and assessments to test students' knowledge and understanding of the concepts learned in the experiments.
- The platform is accessible from anywhere, at any time, making it convenient for students to learn and experiment at their own pace and schedule.
- Virtual Lab is a cost-effective solution for educational institutions, as it eliminates the need for expensive lab equipment and maintenance costs.
- Virtual Lab provides a way for students to learn and experiment in a fun and engaging manner, which can help foster a love for science and technology.

### **To observe the B-H loop of a ferromagnetic material in CRO:**

- A cathode ray oscilloscope (CRO) is an instrument used to display the waveform of electrical signals. It consists of an electron gun, deflection system, and fluorescent screen.
- The B-H loop is a graph depicting the relationship between the magnetic field strength and the magnetic flux density of a ferromagnetic material. It is an important tool for understanding the magnetic properties of a material.
- To observe the B-H loop of a ferromagnetic material in CRO, we need to use a current-carrying solenoid to generate a magnetic field and a test specimen made of ferromagnetic material.
- The current flowing through the solenoid creates a magnetic field inside the test specimen. The magnetic field strength can be varied by adjusting

the current flowing through the solenoid.

- The magnetic flux density in the test specimen can be measured using a Hall probe. The Hall probe is a transducer that converts the magnetic field into a voltage output.
- The output of the Hall probe is connected to the input of the CRO. The CRO displays the waveform of the output voltage from the Hall probe. The horizontal axis of the CRO represents the magnetic field strength, and the vertical axis represents the magnetic flux density.
- By gradually increasing the magnetic field strength, we can observe the B-H loop of the ferromagnetic material. The B-H loop will be displayed on the CRO screen as a closed loop.
- The B-H loop provides information about the magnetic hysteresis of the ferromagnetic material. The area enclosed by the B-H loop represents the energy loss due to hysteresis. The slope of the B-H curve represents the magnetic permeability of the material.
- By analyzing the B-H loop, we can determine the magnetic properties of the ferromagnetic material, such as its coercivity, remanent magnetization, and saturation magnetization.
- In conclusion, observing the B-H loop of a ferromagnetic material in CRO is an important tool for understanding the magnetic properties of materials. The B-H loop provides information about the magnetic hysteresis of the material, which is useful in designing magnetic devices such as transformers, motors, and generators.

### **Introduction to Virtual Lab**

Virtual Lab is an online platform that provides students with the opportunity to perform experiments in a simulated environment. Here are some important points to keep in mind when using Virtual Lab:

### **Benefits of Using Virtual Lab**

- Accessible from anywhere: Virtual Lab can be accessed from any device with an internet connection, allowing students to perform experiments from the comfort of their own homes.
- Cost-effective: Virtual Lab eliminates the need for expensive laboratory equipment and materials, making it a cost-effective alternative to traditional lab experiments.
- Safe: Virtual Lab eliminates the risk of accidents and injuries that can occur during traditional lab experiments.
- Repeatable: Virtual Lab experiments can be repeated as many times as necessary, allowing students to refine their skills and understanding of the experiment.

### **How to Use Virtual Lab**

- Login: First, login to the Virtual Lab website using your username and password.
- Select experiment: Choose the experiment you want to perform from the list of available experiments.
- Read instructions: Read the instructions carefully before starting the experiment.
- Perform experiment: Follow the instructions to perform the experiment.
- Record data: Record your observations and data in the provided space.
- Submit results: Once you have completed the experiment, submit your results and observations.

### **Conclusion**

Virtual Lab is a powerful tool that allows students to perform experiments in a safe and cost-effective environment. By following the instructions carefully and recording your observations accurately, you can improve your understanding of the experiment and learn valuable scientific skills.

### **Determination of the Efficiency of a DC Motor by Loss Summation Method (Swinburne's Test)**

In this experiment, we will learn about the loss summation method, which is also known as Swinburne's test. By using Swinburne's test, we can determine the efficiency of a DC motor. Here are the steps to perform this experiment:

1. Connect the DC motor to be tested with a generator.
2. Now, connect the generator to the supply mains and switch it on.
3. Measure the voltage and current of the generator and record it.
4. Connect a voltmeter across the armature of the DC motor and measure the voltage.
5. Connect an ammeter in series with the DC motor and measure the current.
6. Record the readings of the voltmeter and ammeter.
7. Repeat the above steps for different loads on the motor.
8. Calculate the input power to the motor by multiplying the measured voltage and current.
9. Calculate the output power of the motor by multiplying the measured voltage and current of the armature.

10. Calculate the power loss due to armature resistance by multiplying the square of the armature current with its resistance.
11. Calculate the stray power loss by subtracting the input power and output power from the total power.
12. Calculate the mechanical power output of the motor by subtracting the stray power loss from the output power.
13. Finally, calculate the efficiency of the motor by dividing the mechanical power output by the input power.

By performing this experiment, we can determine the efficiency of a DC motor using the loss summation method or Swinburne's test. This method is widely used in industries to determine the efficiency of DC motors.

### **Course Outcomes:**

Upon completion of this course, students will be able to:

- Understand the fundamental concepts of the subject matter and apply them in practical scenarios.
- Analyze and interpret data using appropriate tools and techniques.
- Design and develop algorithms for solving real-world problems.
- Demonstrate proficiency in programming languages and software development tools.
- Evaluate the performance of algorithms and implement efficient data structures.
- Communicate effectively and work collaboratively in a team environment.
- Apply ethical principles and professional standards in their work.
- Keep pace with emerging trends and technologies in the field.
- Demonstrate the ability to learn and adapt to new technologies and tools.
- Develop critical thinking and problem-solving skills.

These course outcomes are designed to ensure that students are well-equipped with the necessary skills and knowledge to succeed in their future careers. By mastering these outcomes, students will be able to apply their knowledge in a variety of industries and settings, from software development to data analysis and beyond. We encourage students to approach this course with enthusiasm and dedication, and to take advantage of all the resources available to them to achieve these outcomes.

### **Course Outcome (CO) Bloom's Level**

Bloom's level is a framework used to categorize educational goals and objectives. It is a tool used by educators to design assessments, create lesson plans, and evaluate student learning. There are six levels of Bloom's taxonomy: Remembering, Understanding, Applying, Analyzing, Evaluating, and Creating. Each level corresponds to a different cognitive process that students must engage in

to achieve the desired learning outcomes. Below are the course outcomes for each level of Bloom's taxonomy:

### **Remembering**

- Students will be able to recall important concepts, facts, and information related to the course material.
- Students will be able to memorize and recite key terms, definitions, and formulas.
- Students will be able to identify and label important elements within the course material.

### **Understanding**

- Students will be able to explain the meaning of key concepts and ideas.
- Students will be able to summarize important information and ideas from the course material.
- Students will be able to compare and contrast different ideas and concepts within the course material.

### **Applying**

- Students will be able to use their understanding of the course material to solve problems and complete tasks.
- Students will be able to apply course concepts and principles to real-world scenarios.
- Students will be able to demonstrate their ability to use the course material in practical applications.

### **Analyzing**

- Students will be able to break down complex ideas and concepts into smaller parts.
- Students will be able to identify patterns and relationships within the course material.
- Students will be able to evaluate the strengths and weaknesses of different arguments and theories presented in the course material.

### **Evaluating**

- Students will be able to make judgments about the quality and value of the course material.
- Students will be able to evaluate the effectiveness of different strategies and approaches used within the course material.
- Students will be able to critique and analyze different viewpoints presented in the course material.



## Creating

- Students will be able to use their knowledge and understanding of the course material to create new ideas, products, or solutions.
- Students will be able to design and implement original projects and presentations based on the course material.
- Students will be able to generate new ways of thinking about the course material and its applications.

## At the end of this course, the students should be able to:

- Define the basic concepts and principles of the subject matter.
- Understand the historical context and evolution of the subject.
- Analyze and interpret data and information related to the subject.
- Apply the theories and concepts learned to real-world situations and problems.
- Communicate effectively and professionally about the subject matter, both orally and in writing.
- Collaborate with others in group projects and discussions related to the subject.
- Demonstrate critical thinking and problem-solving skills in relation to the subject.
- Evaluate and critique existing research and literature related to the subject.
- Develop a personal perspective and position on the subject, supported by evidence and analysis.
- Engage with the subject matter in a lifelong learning context, keeping up to date with new developments and trends.

By the end of this course, the students will have a strong foundation in the subject matter and the skills necessary to continue learning and growing in their knowledge and understanding of the subject. They will be prepared to apply this knowledge and these skills in various academic and professional contexts, and to engage with the subject matter in a meaningful way throughout their lives.

## **CO 1 Conduct experiments illustrating the application of KVL/KCL and network theorems to DC electrical circuits. K3**

In order to understand the application of KVL/KCL and network theorems to DC electrical circuits, it is important to conduct experiments. Here are some experiments that can be conducted to illustrate the application of these concepts:

1. Experiment to illustrate KVL:
  - Connect a battery, resistor, and voltmeter in series.
  - Measure the voltage across the resistor using the voltmeter.
  - Change the value of the resistor and measure the voltage again.

- Repeat the experiment for different resistor values.
  - Plot the voltage values against the resistor values.
  - The resulting graph should show a linear relationship, which is in accordance with KVL.
2. Experiment to illustrate KCL:
    - Connect a battery, two resistors, and an ammeter in parallel.
    - Measure the current flowing through each resistor using the ammeter.
    - Change the value of one resistor and measure the current again.
    - Repeat the experiment for different resistor values.
    - The sum of the currents flowing through each resistor should be equal to the total current flowing through the circuit, which is in accordance with KCL.
  3. Experiment to illustrate network theorems:
    - Use a circuit with multiple resistors in series and/or parallel.
    - Measure the voltage and current for each resistor.
    - Use Kirchhoff's laws and Ohm's law to calculate the voltage and current for each resistor.
    - Use the superposition theorem to calculate the total voltage and current for the circuit.
    - Use the Thevenin's theorem to calculate the equivalent resistance and voltage of the circuit.
    - Use the Norton's theorem to calculate the equivalent current and resistance of the circuit.

By conducting these experiments, students can gain a better understanding of KVL, KCL, and network theorems and how they are applied to DC electrical circuits.

### **AC Circuits Connected to Single Phase AC Supply**

AC circuits connected to a single phase AC supply exhibit certain behaviors that can be measured and analyzed. Here are some points to keep in mind:

- AC circuits consist of components such as resistors, capacitors, and inductors that affect the behavior of the circuit.
- The voltage and current in AC circuits are constantly changing and can be represented by sinusoidal waveforms.
- The frequency of the AC supply determines the rate at which the voltage and current oscillate.
- The phase angle between the voltage and current waveforms can affect the power factor of the circuit.
- In single phase AC circuits, the voltage and current waveforms have a phase difference of 0 or 180 degrees.
- Power in single phase AC circuits can be measured using techniques such as the wattmeter method or the two wattmeter method.
- The power factor of a single phase AC circuit can be improved by adding capacitors or inductors to the circuit.

### Three Phase Electrical Circuits

In three phase electrical circuits, three AC voltage sources are used instead of one. Here are some points to keep in mind:

- Three phase electrical circuits are more efficient than single phase circuits as they require less conductor material and can deliver more power.
- The voltage waveforms in a three phase system are 120 degrees out of phase with each other.
- The current waveforms in a three phase system are also 120 degrees out of phase with each other.
- The power in a three phase system can be measured using the three wattmeter method or the two wattmeter method.
- The power factor of a three phase system can be improved by adding capacitors or inductors to the circuit.
- Three phase electrical circuits are commonly used in industrial and commercial applications.

### CO 3 Perform experiment illustrating BH curve of magnetic materials. K3

In order to understand the behavior of magnetic materials, it is important to study their B-H or magnetization characteristics. The B-H curve is a graphical representation of the relationship between magnetic field strength (H) and magnetic flux density (B) in a magnetic material. The curve has two parts: the initial magnetization curve (IMC) and the demagnetization curve (DC).

To perform an experiment illustrating the B-H curve of magnetic materials, follow these steps:

1. Obtain a sample of the magnetic material under study. This could be a piece of iron, steel, or any other ferromagnetic material.
2. Set up the experimental apparatus. This will typically involve a magnetizing coil, an ammeter, a voltmeter, and a sample holder.
3. Place the sample in the sample holder and connect the magnetizing coil to a power source.
4. Gradually increase the current in the coil while measuring the magnetic field strength (H) and the magnetic flux density (B) using the ammeter and voltmeter respectively.
5. Plot the values of H and B on a graph to obtain the B-H curve.
6. Determine the values of saturation magnetization (Ms) and coercive force (Hc) from the B-H curve. Ms represents the maximum magnetic flux density that can be induced in the material, while Hc represents the minimum magnetic field strength required to reduce the magnetic flux density to zero.

7. Repeat the experiment for different samples of the same material to obtain an average B-H curve and improve the accuracy of the results.

By performing this experiment, one can gain a better understanding of the magnetic properties of different materials and how they can be used in various applications such as transformers, motors, and generators. It is important to note that the B-H curve may vary depending on factors such as temperature, stress, and magnetic history, and so the experiment should be performed under controlled conditions to obtain accurate results.

#### **CO 4 Calculate Efficiency of a Single Phase Transformer and DC Machine. K4.**

Efficiency is a crucial aspect of electrical machines as it determines how effectively they convert input power into output power. In this topic, we will discuss how to calculate efficiency for a single phase transformer and a DC machine. Here are some key points to go through:

##### **Single Phase Transformer Efficiency Calculation:**

1. The efficiency of a single phase transformer can be calculated using the formula:  $\text{Efficiency} = \text{output power} / \text{input power}$ .
2. The output power of a transformer can be calculated by multiplying the secondary voltage and current ( $P_{\text{out}} = V_{\text{out}} \times I_{\text{out}}$ ).
3. The input power of a transformer can be calculated by multiplying the primary voltage and current ( $P_{\text{in}} = V_{\text{in}} \times I_{\text{in}}$ ).
4. The power losses in a transformer can be calculated by subtracting the output power from the input power ( $P_{\text{l}} = P_{\text{in}} - P_{\text{out}}$ ).
5. The efficiency of a transformer can be calculated by dividing the output power by the input power and multiplying it by 100 ( $\text{Efficiency} = (P_{\text{out}} / P_{\text{in}}) \times 100$ ).

##### **DC Machine Efficiency Calculation:**

1. The efficiency of a DC machine can be calculated using the formula:  $\text{Efficiency} = \text{output power} / \text{input power}$ .
2. The output power of a DC machine can be calculated by multiplying the load voltage and current ( $P_{\text{out}} = V_{\text{load}} \times I_{\text{load}}$ ).
3. The input power of a DC machine can be calculated by multiplying the input voltage and current ( $P_{\text{in}} = V_{\text{in}} \times I_{\text{in}}$ ).
4. The power losses in a DC machine can be calculated by subtracting the output power from the input power ( $P_{\text{l}} = P_{\text{in}} - P_{\text{out}}$ ).

5. The efficiency of a DC machine can be calculated by dividing the output power by the input power and multiplying it by 100 (  $= (P_{out} / P_{in}) \times 100$  ).
6. The efficiency of a DC machine can also be calculated by measuring the input and output power using a wattmeter.

It is essential to understand the efficiency calculation of single phase transformers and DC machines as it helps in determining the performance of these machines. It is also necessary to ensure that these machines are functioning with optimal efficiency to reduce energy consumption and increase their lifespan.

**CO 5 Perform experiments on speed measurement and reversal of direction of three phase induction motor and Identify the type of DC and AC machines based on their construction. K4.**

In this topic, we will be discussing the experiments related to speed measurement and reversal of direction of three-phase induction motor, as well as the identification of DC and AC machines based on their construction. To understand this topic, here are some key points to keep in mind:

- A three-phase induction motor is a type of AC motor that is widely used in industries due to its simple construction and low maintenance cost.
- To measure the speed of a three-phase induction motor, various methods are available, such as the stroboscope method, tachometer method, and slip method.
- The stroboscope method is used to measure the speed of a rotating object by flashing a bright light at regular intervals, creating an illusion of a stationary object.
- The tachometer method uses a device called a tachometer to measure the speed of the motor. It works by measuring the number of revolutions per minute (RPM) of the motor shaft.
- The slip method is used to calculate the speed of the motor by measuring the difference between the synchronous speed of the motor and its actual speed.
- Reversing the direction of a three-phase induction motor is essential in many applications, such as crane operations and conveyor belt systems.
- Reversing the direction of a three-phase induction motor can be achieved by interchanging any two of the three supply lines.
- DC machines are classified into two types: DC generators and DC motors.
- DC generators convert mechanical energy into electrical energy, while DC motors convert electrical energy into mechanical energy.
- AC machines are classified into two types: AC generators and AC motors.
- AC generators convert mechanical energy into electrical energy, while AC motors convert electrical energy into mechanical energy.
- The construction of DC machines is based on Faraday's law of electromagnetic induction, which states that a voltage is induced in a conductor

when it moves through a magnetic field.

- The construction of AC machines is based on the rotating magnetic field theory, which states that a magnetic field is created when three-phase currents flow through three windings placed at an angle of 120 degrees from each other.

By understanding these key points, you will be able to perform experiments on speed measurement and reversal of direction of three-phase induction motor and identify the type of DC and AC machines based on their construction.

#### **K1 – Remember**

- This level requires the student to recall or retrieve previously learned information.
- Techniques such as repetition, mnemonics, and flashcards can help in retaining information.
- Memorization is an essential skill for subjects such as history, language, and sciences.

#### **K2 – Understand**

- This level requires the student to comprehend or explain the meaning of the information.
- Techniques such as note-taking, summarization, and visualization can help in understanding complex concepts.
- Understanding is an essential skill for subjects such as math, literature, and philosophy.

#### **K3 – Apply**

- This level requires the student to use the information in a new or different context.
- Techniques such as problem-solving, experimentation, and simulation can help in applying the learned knowledge.
- Application is an essential skill for subjects such as engineering, business, and computer science.

#### **K4 – Analyze**

- This level requires the student to break down or examine the information to understand its parts and relationships.
- Techniques such as comparison, classification, and deduction can help in analyzing information.
- Analysis is an essential skill for subjects such as law, psychology, and social sciences.

**K5 – Evaluate**

- This level requires the student to make judgments or decisions based on the information and criteria.
- Techniques such as critical thinking, decision-making, and reflection can help in evaluating information.
- Evaluation is an essential skill for subjects such as ethics, politics, and arts.

**K6 – Create**

- This level requires the student to use the information to generate new ideas, products, or solutions.
- Techniques such as brainstorming, design thinking, and prototyping can help in creating new things.
- Creation is an essential skill for subjects such as entrepreneurship, innovation, and arts.